

Lesson 16: Further Aspects of Hypothesis Testing; Relationship between Testing and Interval Estimation

BSTA 551: Statistical Inference

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Lesson 16: Further Aspects of Hypothesis Testing

Review: Where We Left Off

Lesson 15: Neyman-Pearson Lemma; Likelihood Ratio Tests

- The **Neyman-Pearson Lemma** shows the likelihood ratio test is most powerful for simple vs. simple hypotheses
- **UMP tests** exist for one-sided alternatives in exponential families (Karlin-Rubin Theorem)
- The **Likelihood Ratio Test (LRT)** $\Lambda = L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ provides a general recipe for composite hypotheses
- **Wilks' Theorem:** $-2 \ln(\Lambda) \xrightarrow{d} \chi_m^2$ under H_0 (large-sample)
- Classic tests (z, t, F) are all LRTs

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Today's Goals:

1. Understand the distinction between **statistical significance** and **practical/clinical significance**
2. State and use the **duality theorem** connecting confidence intervals and hypothesis tests
3. Construct **confidence intervals by inverting tests** (the general method)
4. Understand **one-sided CIs** and their connection to one-sided tests
5. Use p -value duality and **equivalence testing** in medical research contexts
6. Understand **multiple testing** consequences for CI coverage

Roadmap

Topic	Key Idea	Textbook
Statistical vs. practical significance	A significant result may not be meaningful	Devore 9.6
Duality Theorem	$\theta_0 \in \text{CI}$ iff $H_0 : \theta = \theta_0$ is not rejected	Devore 9.6
Inverting a test	General method for constructing CIs	Devore 9.6
One-sided CIs	Linked to one-sided tests at level α	Devore 9.6
Equivalence testing	Testing that effects are <i>small enough</i>	Applied
Multiple testing & CIs	Simultaneous coverage	Devore 9.6

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Why This Matters

The duality between hypothesis tests and confidence intervals is one of the most elegant unifying results in statistics. It tells us that a confidence interval is **not just a range of plausible values** — it is a compact summary of all hypothesis tests we would fail to reject. In clinical trials, this connection is essential for both planning and reporting.

Statistical Significance vs. Practical Significance (Devore 9.6)

The Core Problem

We have spent much of this course learning how to **detect** an effect. But detection is only half the story.

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💡 Medical Context

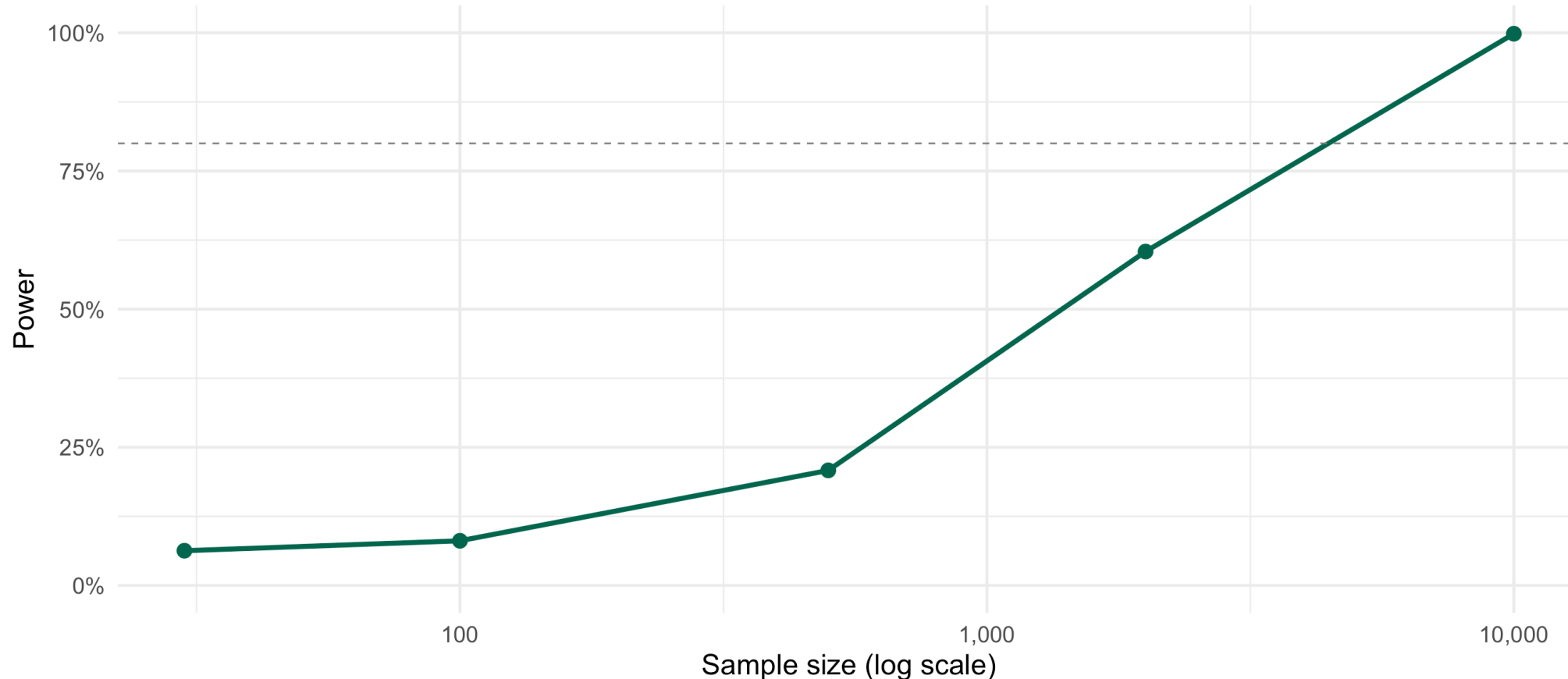
Suppose a new antihypertensive reduces systolic BP by 1 mmHg on average ($n = 50,000, p < 0.0001$). This is statistically significant – but is a 1 mmHg reduction clinically meaningful? The American Heart Association typically considers reductions of ≥ 5 mmHg clinically relevant.

Large Sample Sizes Make Everything Significant

► Code

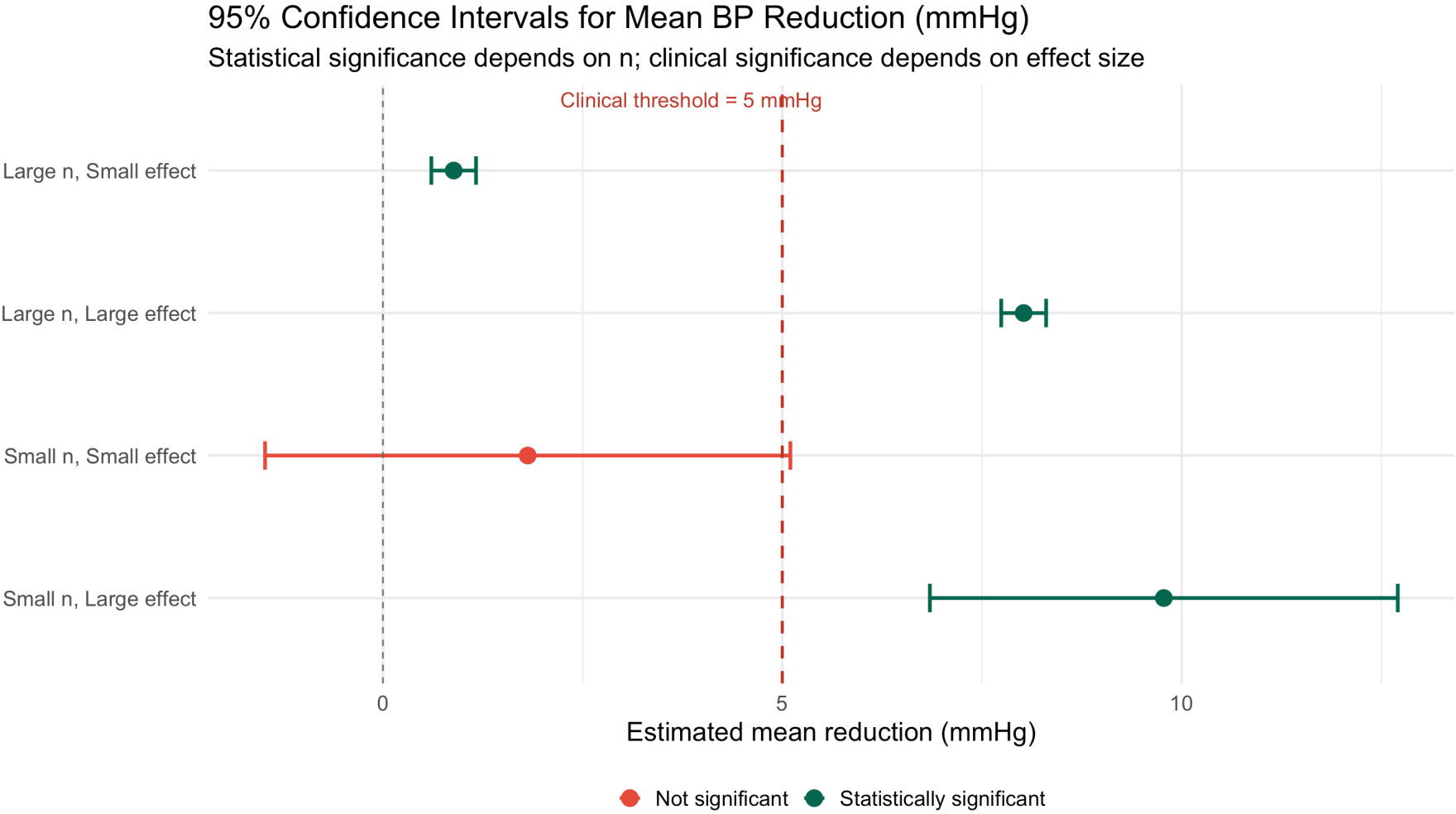
Power to Detect $\delta = 0.5$ mmHg (SD = 10) at $\alpha = 0.05$

A clinically trivial effect becomes detectable with enough patients



Visualizing Statistical vs. Clinical Significance

► Code



Reporting Standards: What Should We Report?

! Best Practice (Devore 9.6 & Medical Journals)

Always report:

1. **The test statistic** (e.g., $t = 3.41$)
2. **The p -value** (e.g., $p = 0.001$)
3. **The confidence interval** for the effect size (e.g., 95% CI: [2.1, 8.7] mmHg)
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Why CIs are essential:

- A CI tells you **the range of plausible effects**, not just whether an effect exists
- A CI directly encodes practical significance: does the entire interval fall below the clinical threshold?
- A narrow CI near zero can rule out clinically meaningful effects even without rejecting H_0

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i Example: CONSORT Guidelines

Major medical journals (NEJM, JAMA, Lancet) following CONSORT guidelines require authors to report effect sizes with CIs, not just p -values. The statement “ $p < 0.05$ ” alone is considered insufficient.

Effect Size Measures

Beyond the raw difference, **effect size** quantifies practical significance on a standardized scale.

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Cohen's d (for means, σ known or estimated):

$$d = \frac{\mu_1 - \mu_2}{\sigma}$$

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< 0.2	Negligible
$0.2 - 0.5$	Small
$0.5 - 0.8$	Medium
> 0.8	Large

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Number Needed to Treat (NNT) for binary outcomes:

$$\text{NNT} = \frac{1}{p_{\text{control}} - p_{\text{treatment}}}$$

Interpretation: How many patients must be treated to prevent one additional event?

Example: Statin Therapy for MI Prevention

A landmark trial finds statins reduce the absolute risk of MI from 4.5% to 3.2% over 5 years.

$$\text{NNT} = \frac{1}{0.045 - 0.032} = \frac{1}{0.013} \approx 77$$

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1 p_control <- 0.045
2 p_treatment <- 0.032
3 RR <- p_treatment / p_control
4 ARR <- p_control - p_treatment
5 NNT <- 1 / ARR
6
7 tibble(
8   Measure = c("Risk (control)", "Risk (treatment)",
9               "Absolute Risk Reduction (ARR)",
10              "Relative Risk (RR)",
11              "Number Needed to Treat (NNT)"),
12   Value = c(p_control, p_treatment, ARR, RR, NNT)
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14 mutate(Value = round(Value, 3)) |>
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Duality of Hypothesis Tests and Confidence Intervals

The Duality Theorem

! Theorem: CI-Test Duality (Devore 9.6)

Let $C(\mathbf{X})$ be a $100(1 - \alpha)\%$ confidence interval for a parameter θ , obtained by inverting a level- α test. Then:

$$\theta_0 \notin C(\mathbf{X}) \iff \text{Reject } H_0 : \theta = \theta_0 \text{ at level } \alpha$$

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💡 Intuition

A 95% CI for μ contains exactly those values of μ_0 for which the data are “consistent” (i.e., would not be rejected at $\alpha = 0.05$).

Proof of Duality

Consider testing $H_0 : \theta = \theta_0$ vs. $H_a : \theta \neq \theta_0$ using a two-sided level- α test.

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$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \theta_0 \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

This is exactly the **95% confidence interval** $\bar{X} \pm z_{\alpha/2} \sigma / \sqrt{n}$!

So: θ_0 is in the CI $\iff$ we fail to reject $H_0 : \theta = \theta_0$. $\square$

The General Method: Inverting a Test

The duality theorem gives us a general method to construct CIs from any test.

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! Definition: Acceptance Region and Confidence Set (Devore 9.6)

For a level- α test of $H_0 : \theta = \theta_0$, the **acceptance region** $A(\theta_0)$ consists of all sample outcomes that do NOT lead to rejection.

The **confidence set** is:

$$C(\mathbf{X}) = \{\theta_0 : \mathbf{X} \in A(\theta_0)\}$$

This set has coverage probability $\geq 1 - \alpha$ for all θ .

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Algorithm for inverting a test:

1. For each candidate value θ_0 , determine the acceptance region $A(\theta_0)$
2. The CI is all values θ_0 for which $\mathbf{X} \in A(\theta_0)$

Example: Inverting the t-Test

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with both μ, σ^2 unknown.

Test $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$ using $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$.

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Acceptance region at level α :

$$A(\mu_0) = \left\{ (x_1, \dots, x_n) : -t_{\alpha/2, n-1} \leq \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \leq t_{\alpha/2, n-1} \right\}$$

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Invert: Find all μ_0 such that $\bar{x}$ falls in $A(\mu_0)$:

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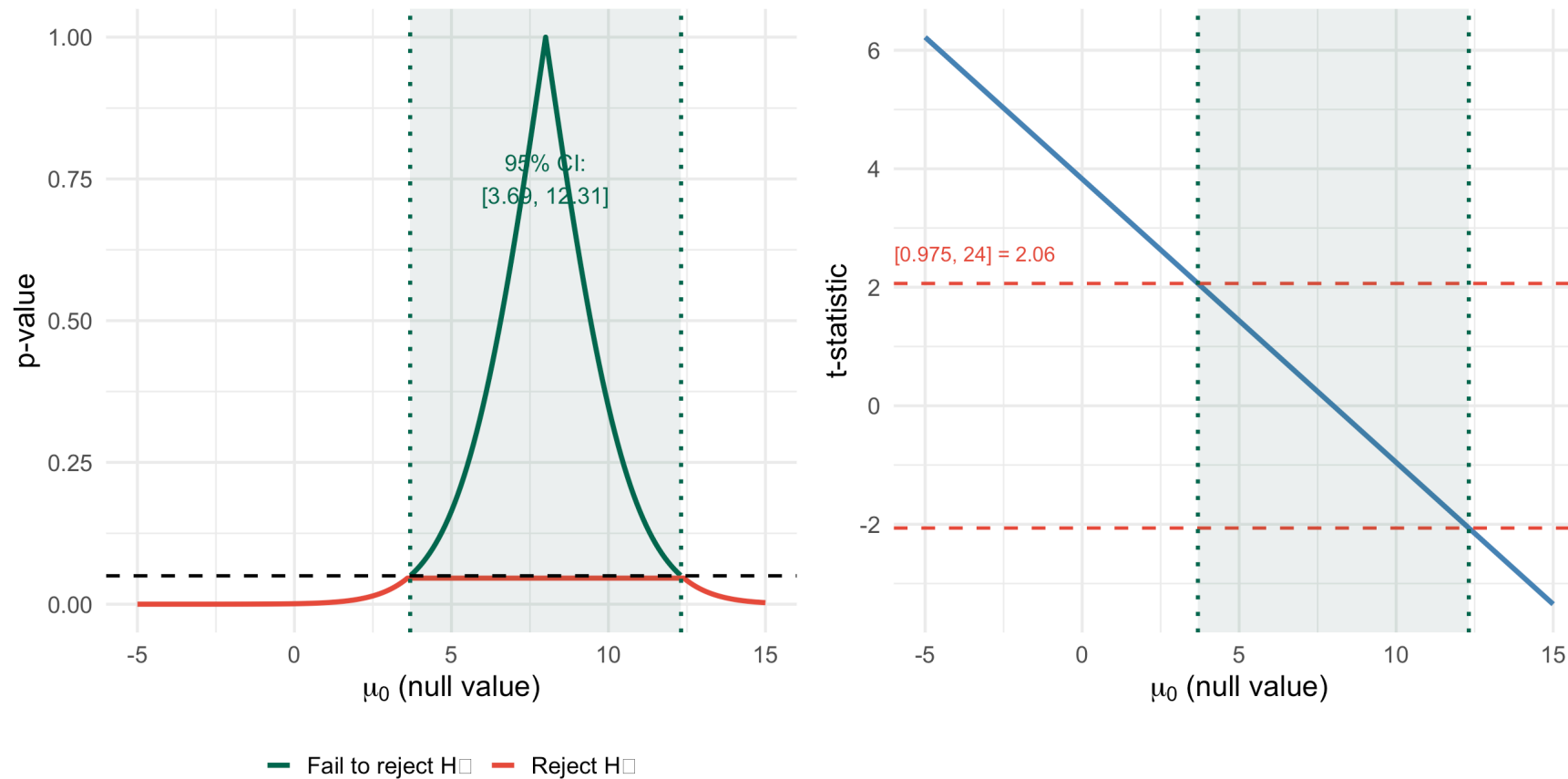
This recovers the familiar **one-sample t confidence interval**. The CI and t-test are **exactly dual** to each other.

Visualizing Duality: Clinical Trial Example

► Code

p-value function for $H_0: \mu = \mu_0$ vs $H_a: \mu > \mu_0$ as a function of null value

Green shaded region = 95% CI = values of μ_0 we fail to reject H_0 Fail to reject when $|t| \leq t_{\text{critical}}$ (green region)



The p-Value Function

The plot on the previous slide illustrates the **p-value function** (or **significance function**):

$$p(\mu_0) = P_{H_0: \mu = \mu_0} (|T| \geq |t_{\text{obs}}|)$$

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- Maximized at the MLE: $p(\bar{x}) = 1$
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Reading the p-Value Function

- The **width** of the region where $p(\mu_0) \geq 0.05$ equals the **width of the 95% CI**
- The **shape** tells us how evidence accumulates against each null value
- This function is more informative than a single p-value or a single CI

One-Sided Tests and One-Sided Confidence Intervals

One-Sided Tests and Bounds

The duality extends naturally to **one-sided** settings.

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! Theorem: One-Sided Duality (Devore 9.6)

Upper-tailed test: $H_0 : \theta \leq \theta_0$ vs $H_a : \theta > \theta_0$ at level α

Reject when $T > c_\alpha$

$\Leftrightarrow$ The corresponding $100(1 - \alpha)\%$ **lower confidence bound** $\underline{\theta}$ satisfies:

$$\theta_0 < \underline{\theta}(\mathbf{X}) \quad \Longleftrightarrow \quad \text{Reject } H_0$$

Lower-tailed test: $H_0 : \theta \geq \theta_0$ vs $H_a : \theta < \theta_0$

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Lower-tailed test: $H_0 : \theta \geq \theta_0$ vs $H_a : \theta < \theta_0$

$\Leftrightarrow$ Upper confidence bound: reject H_0 iff $\theta_0 > \bar{\theta}(\mathbf{X})$

For the one-sample t-test:

Upper bound: $\bar{X} + t_{\alpha, n-1} \cdot S / \sqrt{n}$

Lower bound: $\bar{X} - t_{\alpha, n-1} \cdot S / \sqrt{n}$

Note: uses $t_{\alpha, n-1}$ (one-sided critical value), NOT $t_{\alpha/2, n-1}$

Example: Drug Approval Threshold

A regulatory agency requires a new anticoagulant to reduce stroke risk by **more than 15%** (relative to placebo) to grant approval. We test:

$$H_0 : \text{RR} \geq 1 - 0.15 = 0.85 \quad \text{vs} \quad H_a : \text{RR} < 0.85$$

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```
1 # Observed: 48 strokes in 600 treated, 71 strokes in 600 control
2 n_t <- 600; events_t <- 48
3 n_c <- 600; events_c <- 71
4
5 p_hat_t <- events_t / n_t
6 p_hat_c <- events_c / n_c
7 RR_hat <- p_hat_t / p_hat_c
8
9 # Log-scale test for RR: H0: RR >= 0.85
10 log_RR <- log(RR_hat)
11 se_logRR <- sqrt((1 - p_hat_t)/(events_t) + (1 - p_hat_c)/(events_c))
12
13 # One-sided (lower) test: reject H0 if z < -z_alpha
14 z_stat <- log_RR / se_logRR
15 p_value_onesided <- pnorm(z_stat)
16
17 # 95% upper confidence bound for RR
```

Summary: Two-Sided vs One-Sided Duality

Test Type	Reject H_0 when	Dual CI
Two-sided: $H_a : \theta \neq \theta_0$	$\ \hat{\theta} - \theta_0\ /SE > z_{\alpha/2}$	Two-sided: $\theta_0 \notin [\hat{\theta} \pm z_{\alpha/2} \cdot SE]$
Upper-tailed: $H_a : \theta > \theta_0$	$(\hat{\theta} - \theta_0)/SE > z_\alpha$	Lower bound: $\theta_0 < \hat{\theta} - z_\alpha \cdot SE$
Lower-tailed: $H_a : \theta < \theta_0$	$(\hat{\theta} - \theta_0)/SE < -z_\alpha$	Upper bound: $\theta_0 > \hat{\theta} + z_\alpha \cdot SE$

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Upper-tailed: $H_a : \theta > \theta_0$	$(\hat{\theta} - \theta_0)/SE > z_\alpha$	Lower bound: $\theta_0 < \hat{\theta} - z_\alpha \cdot SE$
Lower-tailed: $H_a : \theta < \theta_0$	$(\hat{\theta} - \theta_0)/SE < -z_\alpha$	Upper bound: $\theta_0 > \hat{\theta} + z_\alpha \cdot SE$

⚠ Common Error

A 95% **two-sided** CI uses $z_{0.025} = 1.96$, while a 95% **one-sided** bound uses $z_{0.05} = 1.645$. These are different! Don't use a two-sided CI to conduct a one-sided test at $\alpha = 0.05$.

Equivalence Testing and Non-Inferiority Trials

Equivalence Testing: The Reverse Problem

Most hypothesis tests ask: **Is there an effect?**

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⚠ Setting

Define a **margin of clinical equivalence** $\delta > 0$.

We want to show the new treatment is **equivalent** to the standard, meaning the true difference $\mu_1 - \mu_2$ falls in $(-\delta, \delta)$.

$$H_0 : |\mu_1 - \mu_2| \geq \delta \quad \text{vs} \quad H_a : |\mu_1 - \mu_2| < \delta$$

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TOST Procedure (Two One-Sided Tests):

Conduct two one-sided tests at level α :

- $H_{01} : \mu_1 - \mu_2 \leq -\delta$ vs $H_{a1} : \mu_1 - \mu_2 > -\delta$
- $H_{02} : \mu_1 - \mu_2 \geq \delta$ vs $H_{a2} : \mu_1 - \mu_2 < \delta$

Conclude equivalence if **both** tests reject.

TOST is Equivalent to a CI Check

! Theorem (Schuirmann 1987)

The TOST procedure at level α rejects both H_{01} and H_{02} if and only if the $100(1 - 2\alpha)\%$ CI for $\mu_1 - \mu_2$ lies entirely within $(-\delta, \delta)$.

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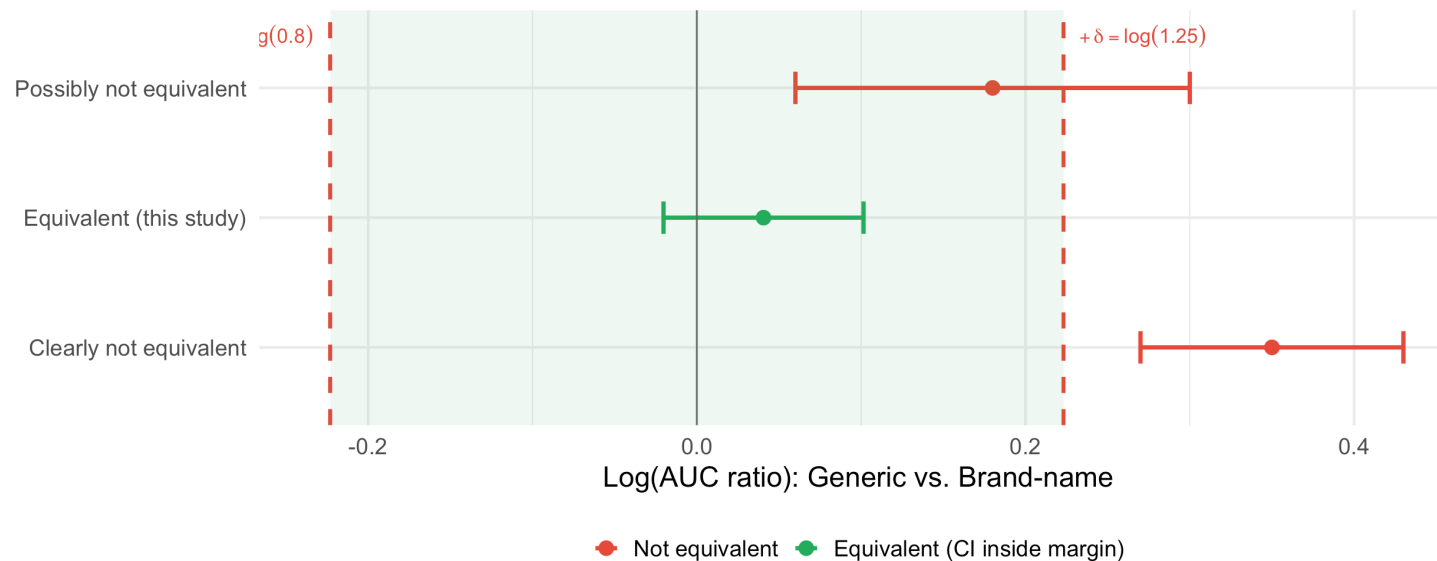
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► Code

Bioequivalence Assessment: 90% CI Must Fall Within Equivalence Margins

FDA standard: $\log(\text{AUC ratio})$ CI within $\log(0.80)$ to $\log(1.25)$



Non-Inferiority Testing

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- The new treatment has fewer side effects or is less costly
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Example

The EINSTEIN-PE trial compared rivaroxaban (once-daily oral) vs. enoxaparin/warfarin for pulmonary embolism. The non-inferiority margin was an HR of 2.0. The upper bound of the 95% CI for the HR was $1.57 < 2.0$, establishing non-inferiority.

Constructing CIs by Inverting Tests: The Pivot Method

The Pivot as the Bridge

The algebraic connection between a CI and a test often works through a **pivotal quantity** — a function of data and parameter whose distribution does not depend on the unknown parameter.

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Key pivots we have used:

Situation	Pivot	Distribution
μ, σ known	$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$
μ, σ unknown	$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$	t_{n-1}
σ^2	$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$	χ^2_{n-1}
Large-sample LRT	$-2 \ln \Lambda$	$\approx \chi^2_m$

Inverting the Chi-Squared Pivot for σ^2

Step 1: The pivot $(n - 1)S^2/\sigma^2 \sim \chi_{n-1}^2$.

Step 2: The acceptance region for testing $H_0 : \sigma^2 = \sigma_0^2$ is:

$$\chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi_{\alpha/2, n-1}^2$$

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$$\frac{(n - 1)S^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma_0^2 \leq \frac{(n - 1)S^2}{\chi_{1-\alpha/2, n-1}^2}$$

This is the **chi-squared CI for σ^2** , derived by inverting the chi-squared test.

Note: $\chi_{\alpha/2, n-1}^2 > \chi_{1-\alpha/2, n-1}^2$, so dividing flips the inequality.

Example: Variability in Drug Concentration

In pharmacokinetics, it's important to control **variability** in drug plasma concentrations. A drug passes QC if $\sigma < 5$ ng/mL (so $\sigma^2 < 25$).

```
1 # 15 measurements of drug plasma concentration (ng/mL)
2 set.seed(551)
3 conc <- rnorm(15, mean = 50, sd = 3.8) # SD ≈ 3.8 (under QC limit)
4
5 n <- length(conc)
6 s2 <- var(conc)
7 df <- n - 1
8
9 # 95% two-sided CI for sigma^2
10 chi_upper <- qchisq(0.975, df) # upper tail => lower bound of CI
11 chi_lower <- qchisq(0.025, df) # lower tail => upper bound of CI
12
13 ci_var_low <- df * s2 / chi_upper
14 ci_var_high <- df * s2 / chi_lower
15 ci_sd_low <- sqrt(ci_var_low)
16 ci_sd_high <- sqrt(ci_var_high)
17
18 cat("Sample variance:", round(s2, 3), "\n")
```

Sample variance: 17.82



Inverting the LRT: Profile Likelihood CIs

For more complex models, we can invert the LRT using Wilks' theorem:

$$C(\mathbf{X}) = \left\{ \theta : -2 \ln \left(\frac{L(\theta)}{L(\hat{\theta})} \right) \leq \chi_{\alpha,1}^2 \right\}$$

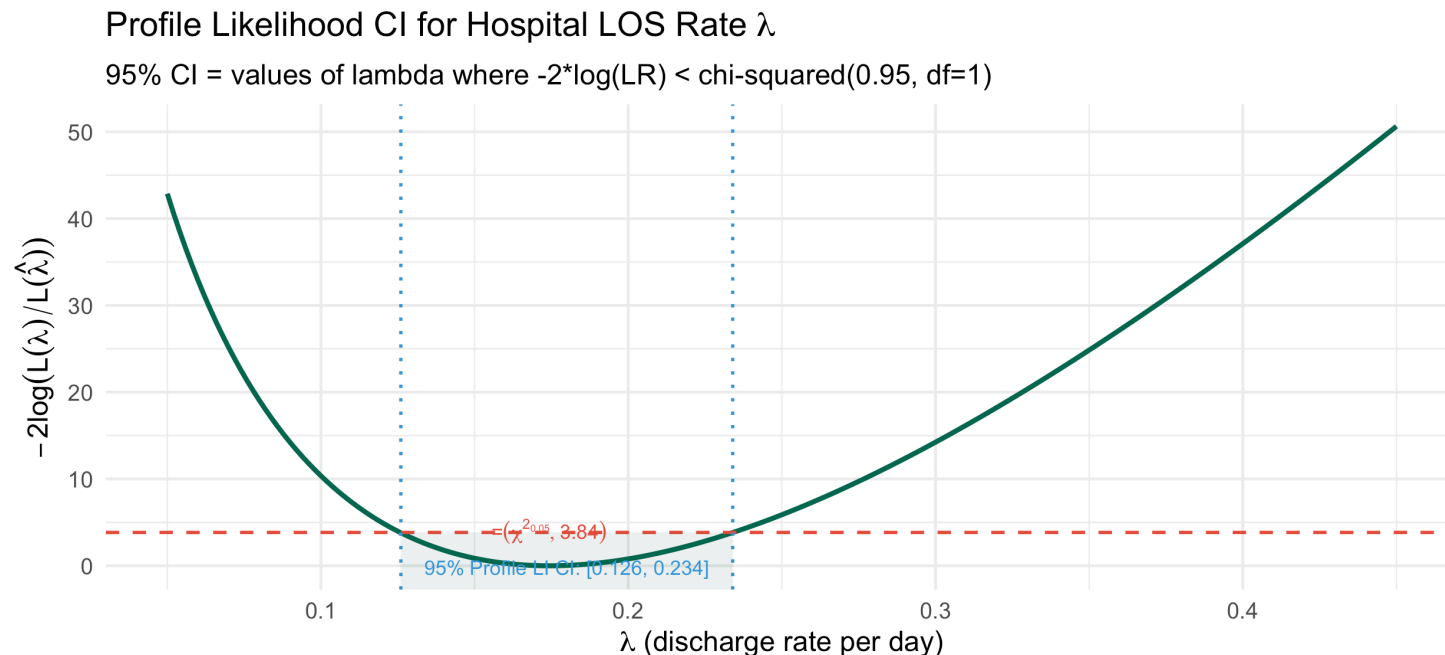
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This is the **profile likelihood confidence interval**, valid for any one-dimensional parameter.

► Code



Multiple Testing and Confidence Intervals

Multiplicity: When We Test Many Hypotheses

When we conduct m tests simultaneously, even if all null hypotheses are true, we expect roughly $m \times \alpha$ false rejections by chance.

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By duality, if we construct m confidence intervals each with coverage $1 - \alpha$:

- Each individual CI has 95% coverage
- But the probability that **all m CIs simultaneously contain their true parameter** is less than 95%

$$P(\text{at least one CI fails}) = 1 - P(\text{all cover}) \leq 1 - (1 - \alpha)^m \approx m\alpha \text{ (Bonferroni)}$$

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! Simultaneous Confidence Intervals

To ensure that **all m CIs simultaneously** contain their true values with probability $\geq 1 - \alpha$:

Use individual level $\alpha^* = \alpha/m$ for each CI (Bonferroni correction).

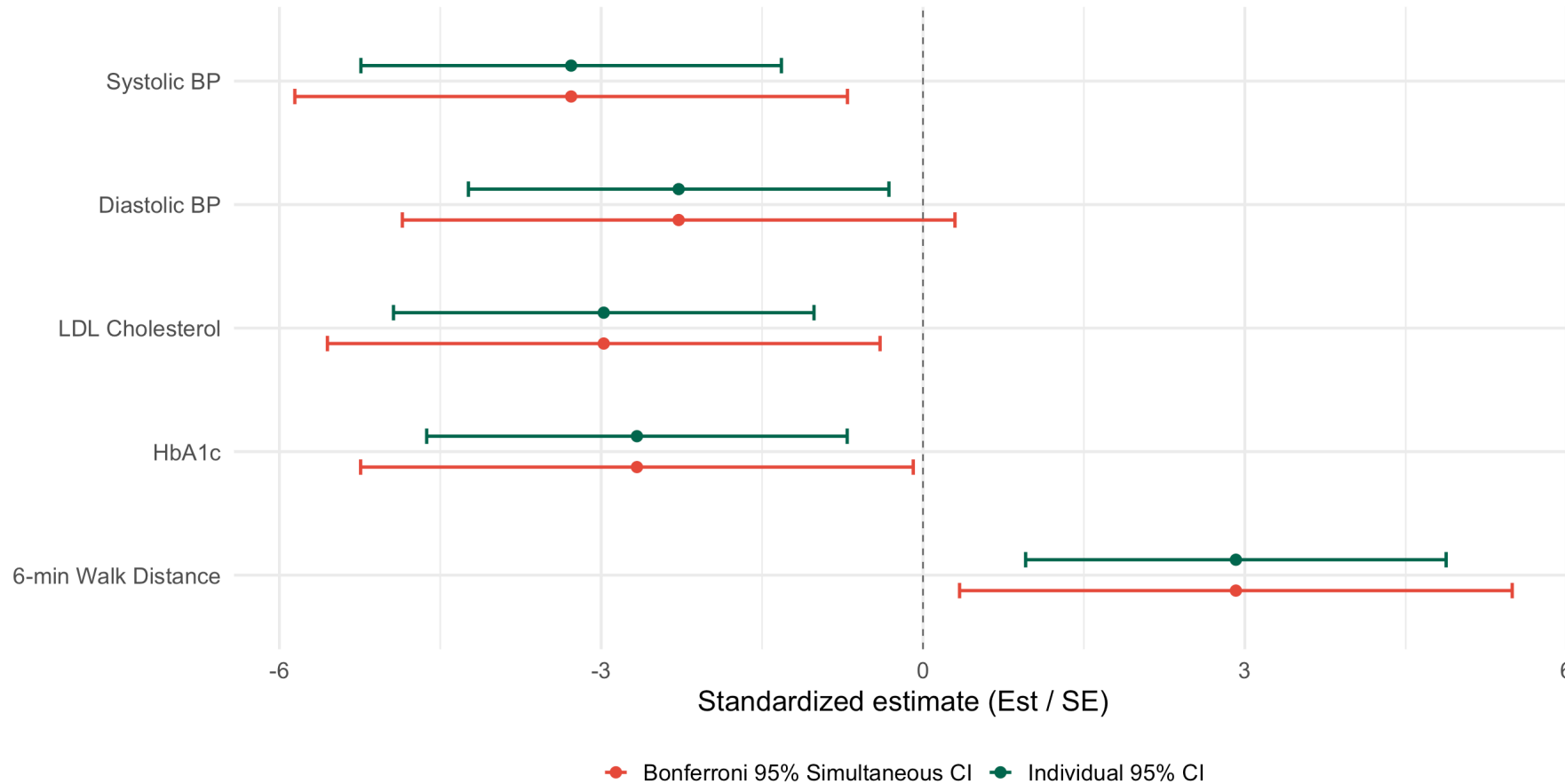
This produces **wider** intervals that account for multiplicity.

Example: Multiple Endpoints in a Clinical Trial

► Code

Individual vs. Bonferroni-Adjusted CIs for 5 Trial Endpoints

Standardized estimates (Est/SE). Bonferroni CIs are wider to control familywise coverage.



Devore 9.6 Textbook Examples

Devore Example 9.6: Testing Hypotheses About a Binomial Proportion

A clinical study examines whether a new screening test has sensitivity greater than 80%.

Let p = true sensitivity. We test $H_0 : p = 0.80$ vs $H_a : p > 0.80$.

In a sample of $n = 200$ subjects with disease, $X = 170$ test positive.

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By exact binomial test and CI:

```
1 # Exact binomial test
2 n_screen <- 200
3 x_pos    <- 170
4
5 binom_test <- binom.test(x_pos, n_screen, p = 0.80, alternative = "greater")
6 binom_test_two <- binom.test(x_pos, n_screen, p = 0.80, alternative = "two.sided")
7
8 cat("One-sided p-value:", round(binom_test$p.value, 5), "\n")
```

One-sided p-value: 0.04302

```
1 cat("95% lower bound for p:", round(binom_test$conf.int[1], 4), "\n")
```

95% lower bound for p: 0.8021

```
1 cat("\nTwo-sided 95% CI for p: [",
2     round(binom_test_two$conf.int[1], 4), ",",
3     round(binom_test_two$conf.int[2], 4), "]\n")
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Two-sided 95% CI for p: [0.7078, 0.8965]



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CI-Test Duality Illustrated (Textbook Example)

Consider $X \sim \text{Poisson}(\mu)$. We observe $X = 3$ and wish to test $H_0 : \mu = 5$ vs $H_a : \mu < 5$.

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1 # Exact Poisson test
2 x_obs <- 3
3 mu_0 <- 5.0
4
5 # One-sided: lower-tailed (alternative: mu < 5)
6 p_lower <- ppois(x_obs, lambda = mu_0)
7 cat("P(X <= 3 | mu = 5):", round(p_lower, 5), "\n\n")
```

P(X <= 3 | mu = 5): 0.26503

```
1 # 95% two-sided CI for mu via exact Poisson intervals
2 # Lower bound: qchisq(0.025, 2*x) / 2
3 # Upper bound: qchisq(0.975, 2*(x+1)) / 2
4 ci_low_pois <- qchisq(0.025, 2 * x_obs) / 2
5 ci_high_pois <- qchisq(0.975, 2 * (x_obs + 1)) / 2
6
7 cat("Exact 95% CI for mu: [", round(ci_low_pois, 4), ",",
8     round(ci_high_pois, 4), "]\n")
```

Exact 95% CI for mu: [0.6187 , 8.7673]

```
1 cat("mu_0 = 5 in CI?", ci_low_pois <= mu_0 & mu_0 <= ci_high_pois)
```

mu_0 = 5 in CI? TRUE

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Key Takeaways and Looking Ahead

Key Takeaways

! Today's Main Points

1. **Statistical significance $\neq$ practical/clinical significance:** always report effect sizes and CIs alongside p-values
2. **CI-Test Duality (Devore 9.6):** $\theta_0 \in \text{CI}(X) \iff$ fail to reject $H_0 : \theta = \theta_0$ at the same α
3. **Inverting a test** provides a general, principled method for constructing CIs from any hypothesis test
4. **One-sided duality:** a $100(1 - \alpha)\%$ one-sided bound corresponds to a one-sided test at level α (using z_α , not $z_{\alpha/2}$)
5. **Equivalence/non-inferiority testing:** FDA bioequivalence uses 90% CI (TOST at $\alpha = 0.05$); non-inferiority uses one-sided lower bound
6. **Multiple testing:** simultaneous CIs must be widened (e.g., Bonferroni) to maintain joint coverage
7. **Profile likelihood CIs** invert the LRT using Wilks' theorem: a general tool for any parameter

Looking Ahead

Next class (Lesson 17): Bootstrap Hypothesis Testing (C&H Chapter 8)

- The bootstrap hypothesis test: resampling under H_0
- One-sample and two-sample bootstrap tests
- Permutation tests: exact null distribution by permuting labels
- Comparison: parametric vs. bootstrap vs. permutation tests
- Applications to clinical trial data

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 9.6.
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- Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680.
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: context, process, and purpose. *The American Statistician*, 70(2), 129–133.

Questions?

Thank you!